HYPERTENSION CARDIOLOGY CENTRE ANALYTICS REPORT

Improving Blood Pressure Control: Predictors and Treatment Optimization

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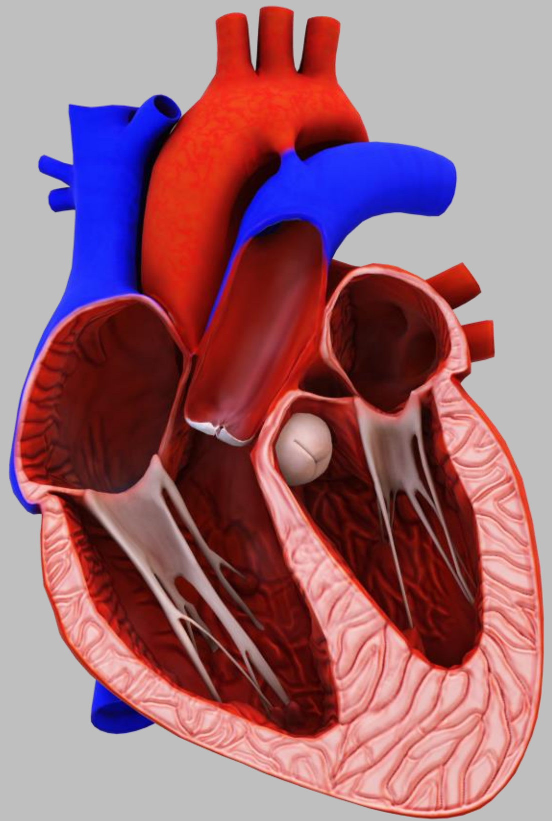

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What this report covers

Data Source & Methodology

Dataset Profile

Records:	350 patient entries
Original columns:	16 variables
New columns added:	4 engineered features
Total columns:	20 after processing
Duplicates found:	0 — dataset is clean
Date range:	2024 cohort
Source:	Hypertension Cardiology Centre Records

4 Engineered Features

Age Group

Categorised continuous age into:
Under 45 · 46–55 · 56–65 · 66+

Comorbidity Rate

Count of active comorbidities (0–4)
per patient for burden scoring

BP Level Before Prescription

Clinical staging at baseline:
Stage 2 Hypertension · Hypertensive Crisis

BP Level After Medication

Post-treatment BP classification for transition & outcome analysis
Normal, Elevated, Isolated Diastolic Hypertension, Stage 1 Hypertension, Stage 2 Hypertension, Hypertension Crisis

Dataset Overview

2024 cohort

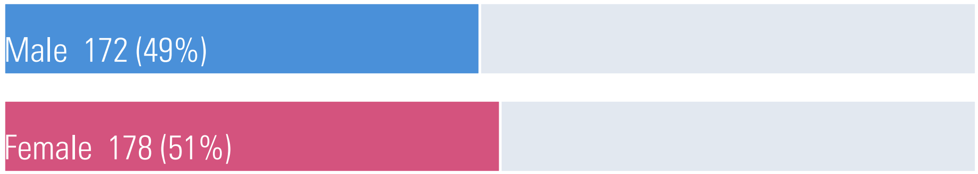
350
Total Patients
Health records

57.4
Mean Age (yrs)
Range: 30–85

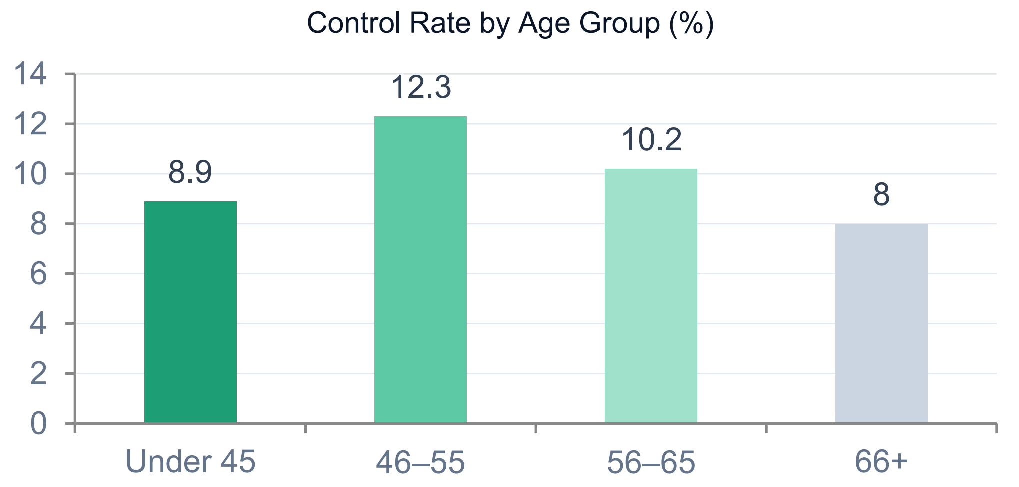
21.1%
Best Drug Class Combination control rate
12 of 57 patient usage

9.4%
BP Control Rate
33 of 350 patients

Gender Distribution



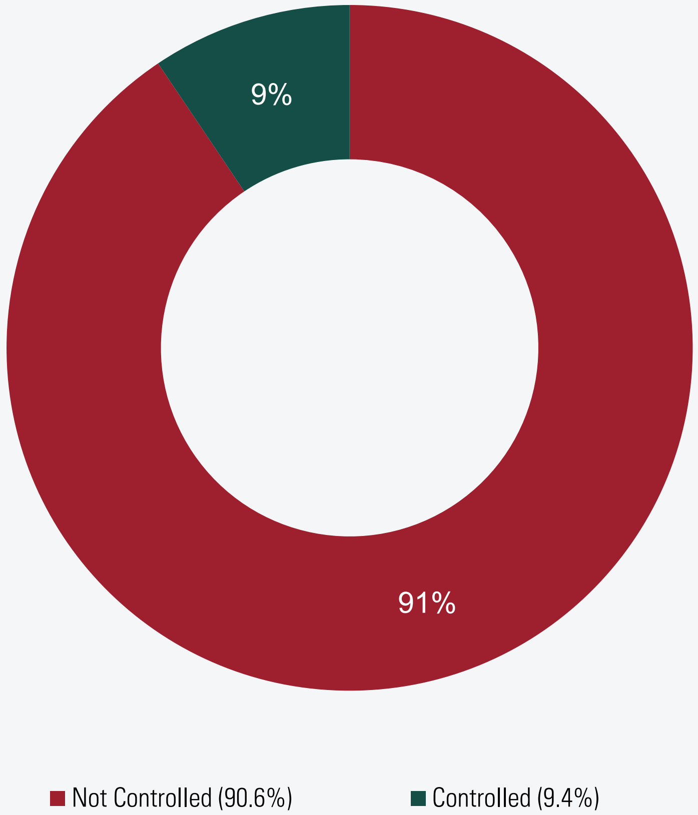
Near-equal gender split



BP Control Outcomes at 3 Months

Overall performance and key driver summary

Control Status (N=350)



Top Predictors of BP Control Failure

01 **Medication Adherence**

Poor adherence is seen as the strongest predictor of failure

02 **BP Stage at Prescription**

Hypertensive Crisis: 0% control across ALL drug classes
Stage 2 Hypertension: 12.5% control across ALL drug classes
Because baseline strongly has higher systolic and diastolic bp, they strongly predict failure

03 **Drug Class Selection**

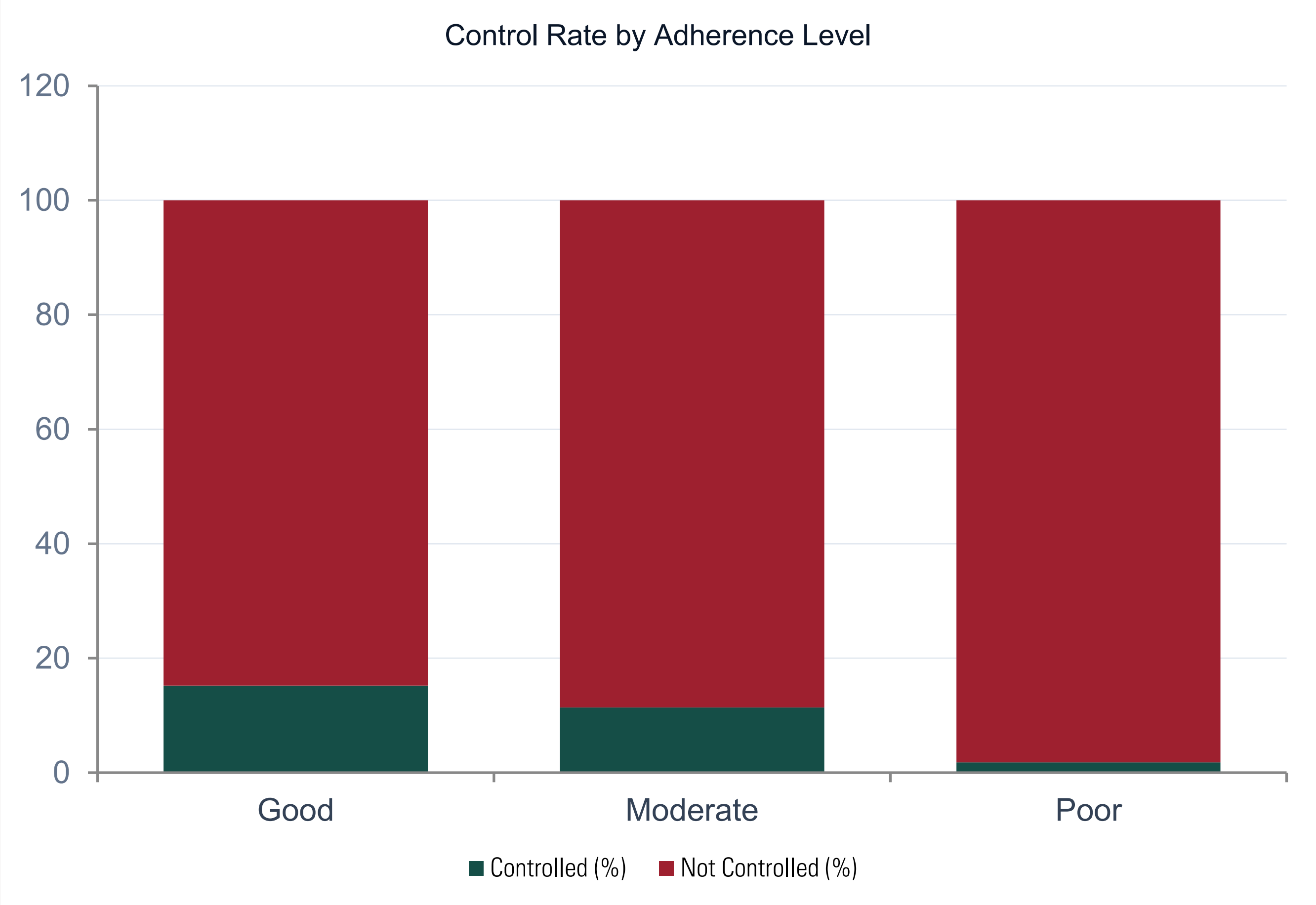
Combination Therapy: 21.1% control — only class above 13%

04 **Visit Frequency**

Patients with ≤1 visit had 5.3% control vs 15.9% for 2 visits

Medication Adherence

The single strongest modifiable predictor at point of prescription



15.2% 105 patients (30% of total patients)

Good adherence control rate

11.4% 132 patients (38% of total patients)

Moderate adherence control rate

1.8% 113 patients (32% of total patients)

Poor adherence control rate

8× Good vs Poor adherence patients

Control gap

BP Stage at Prescription

Hypertensive Crisis patients achieved 0% control across all drug classes

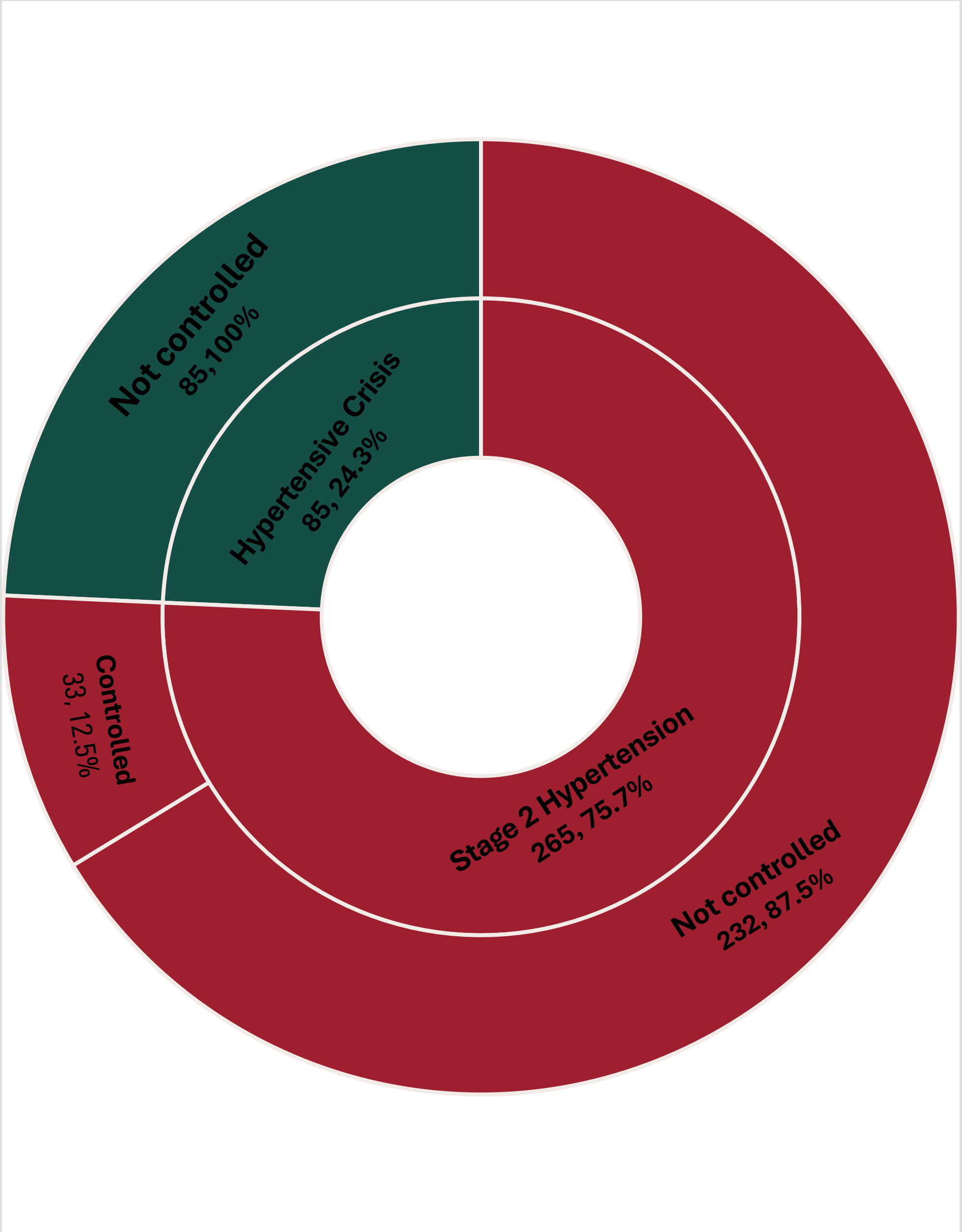
12.5%
Stage 2 Hypertension
265 patients

0.0%
Hypertensive Crisis
85 patients

BP Stage Transition: Before → After Medication

BP Stage After	Normal	Elevated	Isolated Diastolic	Stage 1	Stage 2	Crisis
Stage 2 HTN (n=265)	7	29	6	40	183	0
Hypertensive Crisis (n=85)	0	0	0	0	74	11

High Risk Patient at Point of Prescription



Antihypertensive Drug Class Performance

Control rate and systolic BP reduction across 6 drug classes

Drug Class Against Control Success Rate

Antihypertensive Class	0	1	2	3	4	Total Cases Used
ACE Inhibitor						**45**
ARB		1	2	1		**68**
Beta Blocker			3	3		**61**
Calcium Channel Blocker	1		1	1		**58**
Combination Therapy		4	3	5		**57**
Thiazide Diuretic	2		2	2	2	**61**

Four lab-test were carried out

Comorbidity Profile
0 – All Lab-test tested negative
1 – One of the lab-test tested positive
2 – Two of the lab-test tested positive
3 – Three of the lab-test tested positive
4 – All the lab-test tested positive

From the table above, Combination therapy delivers the best outcome for patients that has the positive of either one, two, three of the underlying factors amongst other Antihypertensive drugs

ACE Inhibitor shows a control rate of 0% across 45 patients, making it the least effective of the drugs

Drug Class Against Control Success Rate

Antihypertensive Class	0	1	2	3	4
ACE Inhibitor					
ARB		1	2	1	
Beta Blocker			3	3	
Calcium Channel Blocker	1		1	1	
Combination Therapy		4	3	5	
Thiazide Diuretic	2		2	2	2

Drug Class Against Control Failure Rates

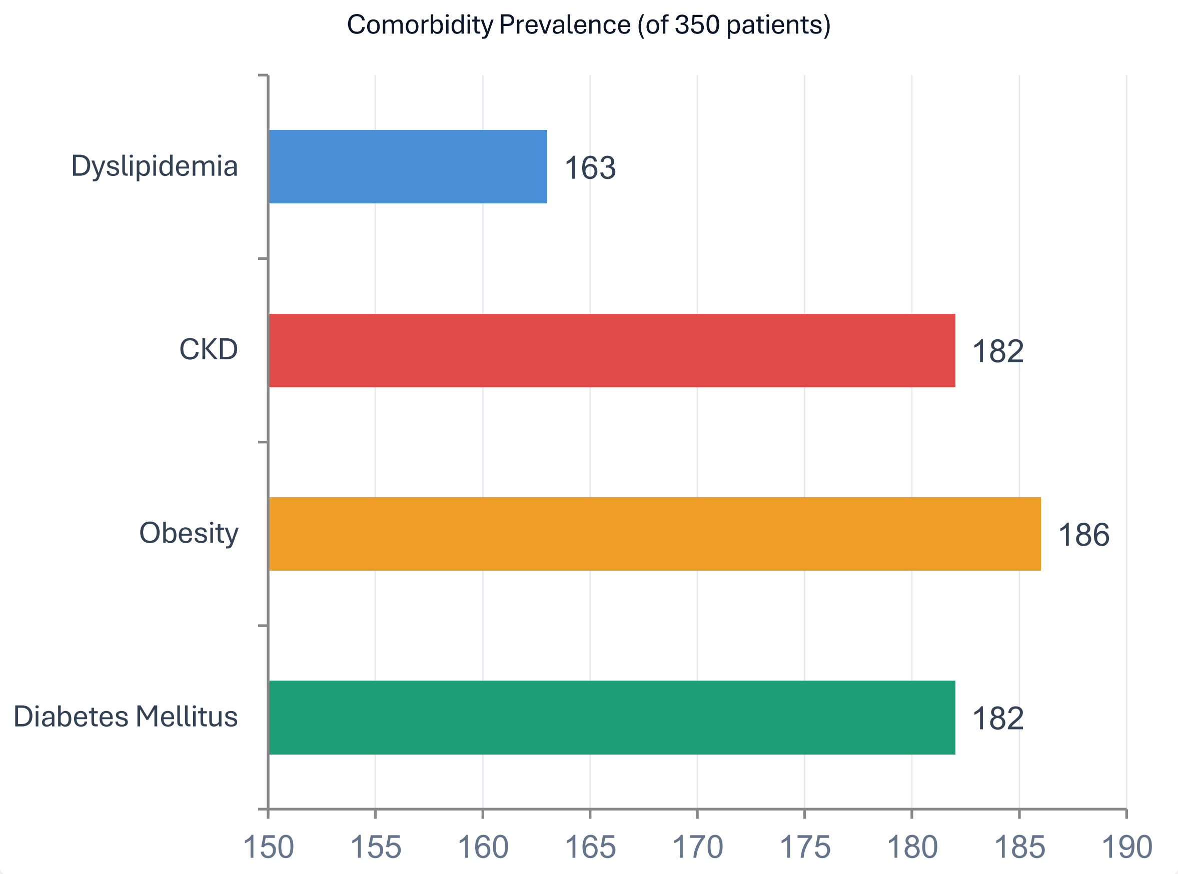
Antihypertensive Class	0	1	2	3	4
ACE Inhibitor	1	14	15	13	2
ARB	7	18	17	19	3
Beta Blocker	1	17	14	18	5
Calcium Channel Blocker	4	12	23	14	2
Combination Therapy		7	24	13	1
Thiazide Diuretic	1	15	20	12	5

The two table shows an extensive correlation of Drug classes and their control rates with the comorbidity profile.

Zero comorbidity profile interprets no undelaying factor is found positive but leaves us with a question as to why 99.8% of the patients found in that group are still hypertensive. Further analysis is required to be done

Comorbidity Distribution

High prevalence of multiple comorbidities complicates treatment



Key Comorbidity Insights

- 52% of patients have Diabetes Mellitus — requiring RAAS-protective drug selection (ARB preferred)
- 53% have Obesity — metabolic resistance increases treatment complexity
- 52% have CKD — ACE Inhibitor showed 0% control despite guideline preference; monitor eGFR
- 47% of patients have Dyslipidemia

Insights

Problem 1: What predicts BP control failure at 3 months?

- **Medication Adherence**

Across all 350 patients, medication adherence showed the clearest separation in outcomes. With 32% of the records rated as Poor and 38% as Moderate, only 30% of patients entered the treatment period with a behavioral profile conducive to success. This means the majority of control failures were predictable before a single dose was dispensed.

- **Finding 2: BP Stage at Prescription**

Patients presenting in Hypertensive Crisis (85 patients) achieved **0% control** across every single drug class. Not one patient in this group reached the controlled threshold at 3 months, regardless of what they were prescribed or their adherence level. Patients in Stage 2 Hypertension fared modestly better at 12.5%, but the crisis subgroup represents a fundamentally different clinical scenario.

Insights

Problem 2 : Which drug class delivers the best outcomes by comorbidity profile?

- **Combination Therapy**

Across the full dataset, Combination Therapy achieved the highest control rate at **21.1%**. It outperformed every single agent across most comorbidity profiles and is the only drug class that meaningfully bridges the challenge of high baseline BP. The next best single agent, Thiazide Diuretic, trails at 13.1%.

- **The ACE Inhibitor paradox**

ACE Inhibitor produced the second highest average systolic drop in the dataset (15.9 mmHg) but recorded **0% control** across all 45 patients prescribed it. This pattern — high BP reduction with no controlled outcomes — suggests that this drug is being prescribed to patients with very high baseline BP who respond to the drug but cannot reach the controlled threshold from where they start. This could be a patient selection problem, or a drug efficacy problem, and it calls for a further analysis.

Insights Cont’d

- **The zero-comorbidity observation — an open clinical question**

The data shows that patients with no documented comorbidity (no Diabetes, no CKD, no Dyslipidemia, no Obesity) still present with hypertension requiring treatment, and 99.8% of them remain uncontrolled at 3 months despite. This is clinically paradoxical. A patient with no identifiable comorbidity burden would typically be expected to have the most treatment-responsive form of hypertension. Yet the records-wide failure rate suggests that the absence of documented comorbidities does not mean the absence of underlying aetiology. The clinical recommendation is that zero-comorbidity patients who remain uncontrolled at 3 months should not simply be escalated pharmacologically. They require a structured secondary investigation to identify whether their hypertension has a treatable underlying cause that current treatment is not addressing. This is a gap the current dataset cannot close, it requires a prospective study with expanded variable capture.

Insights Cont’d

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Recommendation

1. Prioritize adherence-focused interventions

- Implement patient education programs
- Use reminders, follow-ups, or digital adherence tools
- Simplify medication regimens where possible

2. Intensify treatment for high-risk patients early

For patients with: SBP ≥160 mmHg and Multiple comorbidities. Consider:

- Early combination therapy
- Closer clinical monitoring
- More aggressive management strategies

3. Focus on quality, not just frequency of visits

Since visit timing had little effect: Emphasize what happens during the visit, not just how often

Include:

- Medication counseling
- Lifestyle guidance
- Adherence checks

Glossary

- **Diabetes Mellitus:** A chronic metabolic disorder characterized by high blood sugar (hyperglycemia) resulting from the body's inability to produce enough insulin or use it effectively.

- **Chronic Kidney Disease (CKD):** A condition marked by the long-term (more than 3 months) loss of kidney function or structural damage, often measured by a reduced glomerular filtration rate (eGFR <60 ml/min/1.73m2) or high albumin (protein) in the urine.

- **Dyslipidemia:** An unhealthy imbalance of lipids (fats) in the blood. This typically includes high levels of triglycerides and "bad" LDL cholesterol, combined with low levels of "good" HDL cholesterol.

- **Obesity:** A chronic, progressive disease defined by an excessive accumulation of body fat that poses health risks.

- **Comorbidity** is a medical term used to describe the coexistence of two or more chronic conditions in the same person
- **Control rate** is the percentage of people with a specific condition who have successfully reached their clinical targets